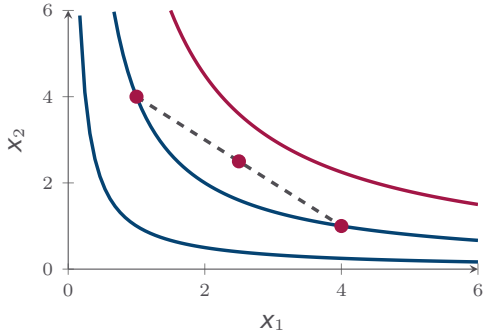
Cobb–Douglas: mixing can help



$$u(x) = \sqrt{x_1 x_2}.$$

$u(1,4) = u(4,1) = 2$, but

$$u(2.5, 2.5) = 2.5.$$

On $\mathbb{R}^2_{++}$: strongly monotone and strictly convex preferences.
On $\mathbb{R}^2_{+}$, axis bundles have zero utility; the strict claims fail at the boundary.